This document provides a more in-depth summary of the methodologies, results, and possible extensions of my method for evaluating a pass rusher's skill. I used NFL player tracking data from the 2025 Big Data Bowl to accomplish my main goal: to create a context-adjusted pressure rate and technique profile for each pass rusher by using features from the tracking data, classifying how each pass rush rep was won, and adjusting for the difficulty of blocking assignments that were faced (single blocker vs. double-teamed).

Data and Preprocessing

I filtered the dataset to passing situations only by keeping dropbacks from the plays dataset. Pass rushers were identified using the 'wasInitialPassRusher' flag in player_play.csv. To focus on EDGE players, I limited the sample to defensive ends and outside linebackers and applied a weight filter of under 300 pounds to remove likely interior defensive linemen that had loose roster labels.

All tracking coordinates were standardized so the offense always moved in the positive x-direction. This made plays comparable regardless of whether the original play direction was left or right. After filtering, the final dataset included 23,345 pass rush attempts from 253 unique rushers.

Feature Engineering

Two features captured how quickly a rusher started the rep. The first was 'getOffTimeAsPassRusher', which was provided in the dataset and measures how quickly the rusher crosses the line of scrimmage. The average get-off time was 0.91 seconds. The second was peak burst speed, calculated as the rusher's maximum speed during the first 1.5 seconds after the snap. The average peak burst speed was 3.9 yards per second.

Each pass rush attempt was also assigned a blocking context using the 'blockedPlayerNFLId' columns in player_play.csv, which identifies the blockers assigned to each rusher. Rush attempts were labeled as unblocked, single-blocked, or double-teamed. The distribution was 67% of rush attempts were single-blocked, 24% double-teamed, and 9% unblocked.

To describe the actual rush attempt, I calculated 5 tracking-based metrics from the rusher and primary blocker positions post-snap: separation gained at 1.5 seconds, separation gained at 2.5 seconds, blocker displacement as an indicator of power, rusher direction change between 1.0 and 2.0 seconds as an indicator of a counter move, and blocker orientation change at 1.5 seconds as a measure of how much the blocker had to react. Thresholds were calibrated by comparing pressure and non-pressure plays within single-blocked pass rush attempts.

Win Type Classification

Each of the 20,080 **blocked** rush attempts was assigned one win type using a priority cascade: **Counter, Power, Speed,** or **Loss**. The resulting distribution was **Loss: 54%, Speed: 40%, Power: 3%, and Counter: 3%.**

The main takeaway is that most wins in this framework come from creating separation with speed. Power and counter wins appear much less often at the individual rep level, which makes sense because they require more specific movement patterns and are harder to isolate cleanly just from tracking data alone.

Context-Adjusted Pressure Rate

Raw pressure rate tends to be misleading because it does not account for the blocking assignment a rusher faced. A pressure against a double-team should be valued differently than a pressure against a single blocker, when evaluating an individual pass rusher. To adjust for this, each rush attempt was scored as the rusher's pressure outcome divided by the league-average pressure rate for that blocking context.

The baselines were calculated directly from the data. Single-blocked rushes generated pressure at **11.8%,** while double-teamed rushes generated pressure at **7.1%.** A player's context-adjusted pressure rate is the average of these scores across all attempts. A score of **1.0** represents league average, while a score of **2.0** means the rusher generated pressure at twice the expected rate given the blocking looks faced.

Model Validation and Results

To validate the tracking features, I used a logistic regression model as a simple check rather than as the final player-ranking method. The goal was to see whether the movement features could help separate pressure reps from non-pressure reps.

I trained the model on single-blocked rushes to predict whether a rush created pressure. Single-blocked reps were used because they provide the cleanest one-on-one matchup between a rusher and a blocker. This makes it easier to connect the rusher's movement and blocker interaction to the final pressure outcome.

The model was tested by splitting the data into five different train-test groups, so it was evaluated on plays it had not already seen. It produced a **ROC-AUC score of 0.623**. ROC-AUC measures how well the model separates pressure reps from non-pressure reps. A score of **0.50** is roughly the same as random guessing, so a score above **0.60** showed that the features had real predictive value, even though they did not explain every pressure outcome. Blocker displacement was the strongest positive predictor, suggesting that how much a rusher moves the blocker is a useful indicator of power on a pass rush rep.

As another validation check, all four consensus elite rushers in the sample, **Myles Garrett, Nick Bosa, Trey Hendrickson, and Micah Parsons**, ranked in the top 20% of the **108 qualified players** with at least 75 rush attempts.

Myles Garrett led all qualified rushers with a context-adjusted pressure rate of **2.10**, followed by **Za'Darius Smith** at **1.83**, **Justin Houston** at **1.82**, **Trey Hendrickson** at **1.77**, and **Nick Bosa** at **1.77**. Rushers with a double-team rate above 25% were flagged as scheme priorities because offenses appeared to devote extra protection resources toward them. The technique profiles also produced useful takeaways: Hendrickson had the highest power index among qualified rushers at **5.01**, while Parsons ranked sixth overall with a more balanced profile across speed, power, and counter wins.

Limitations and Future Extensions

This approach has a few important limitations. The weight filter removes obvious roster-label issues, but some lighter interior defenders can still pass through as EDGE rushers. The win type thresholds were calibrated from pressure and non-pressure distributions, but they were not fully optimized against an outcome variable. The validation model was also trained only on single-blocked reps. This was intentional because single-blocked plays give the cleanest view of the rusher-blocker matchup, while double-teamed reps introduce more noise from help blocks, protection slides, chips, and scheme design. However, this means the model validation does not fully explain pressure creation in more complex blocking situations. The dataset also covers only nine weeks of one season, so rankings near the 75-attempt minimum may be sensitive to sample size. Finally, while a ROC-AUC of 0.623 confirms that the features have signal, a large amount of pressure variance remains unexplained.

Future work could improve the method by using pre-snap alignment, especially y-coordinate and distance from the offensive tackle, to identify true EDGE rushers instead of relying on roster position and weight. The analysis could also be repeated across multiple seasons to measure year-over-year stability, separate repeatable skill from short-term noise, and improve confidence in player rankings. Additional extensions include incorporating blocker athleticism, replacing discrete frame snapshots with full-trajectory models, and aggregating win type profiles at the team level to study scheme fit and roster construction needs.